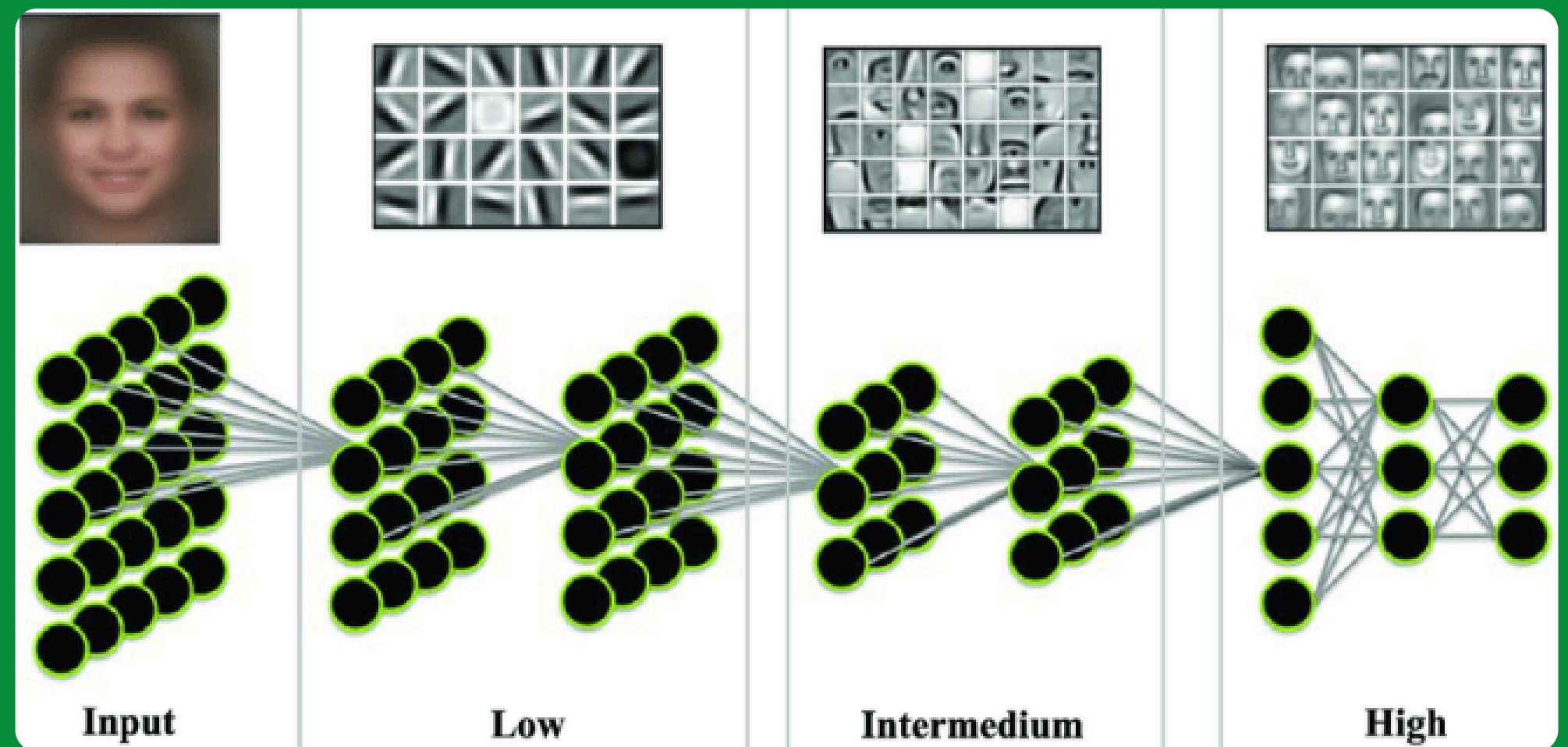


AND WHAT DOES IT TELL US ABOUT OURSELVES



Echobox

TALK THAT
SCIENCE

Who are we?

TALK THAT
SCIENCE



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Talk That Science

Amity

Talk That Science, a monthly talk show in which we play the nicest tunes while talking about the coolest research. We invite both scientists and students to open up a discussion.

Talk/Informative

NEXT SHOW

Saturday, 06/06/2026



Who is Peter Bloem?



Who is Micha Heilbron?



Machine learning ? Artificial Intelligence?

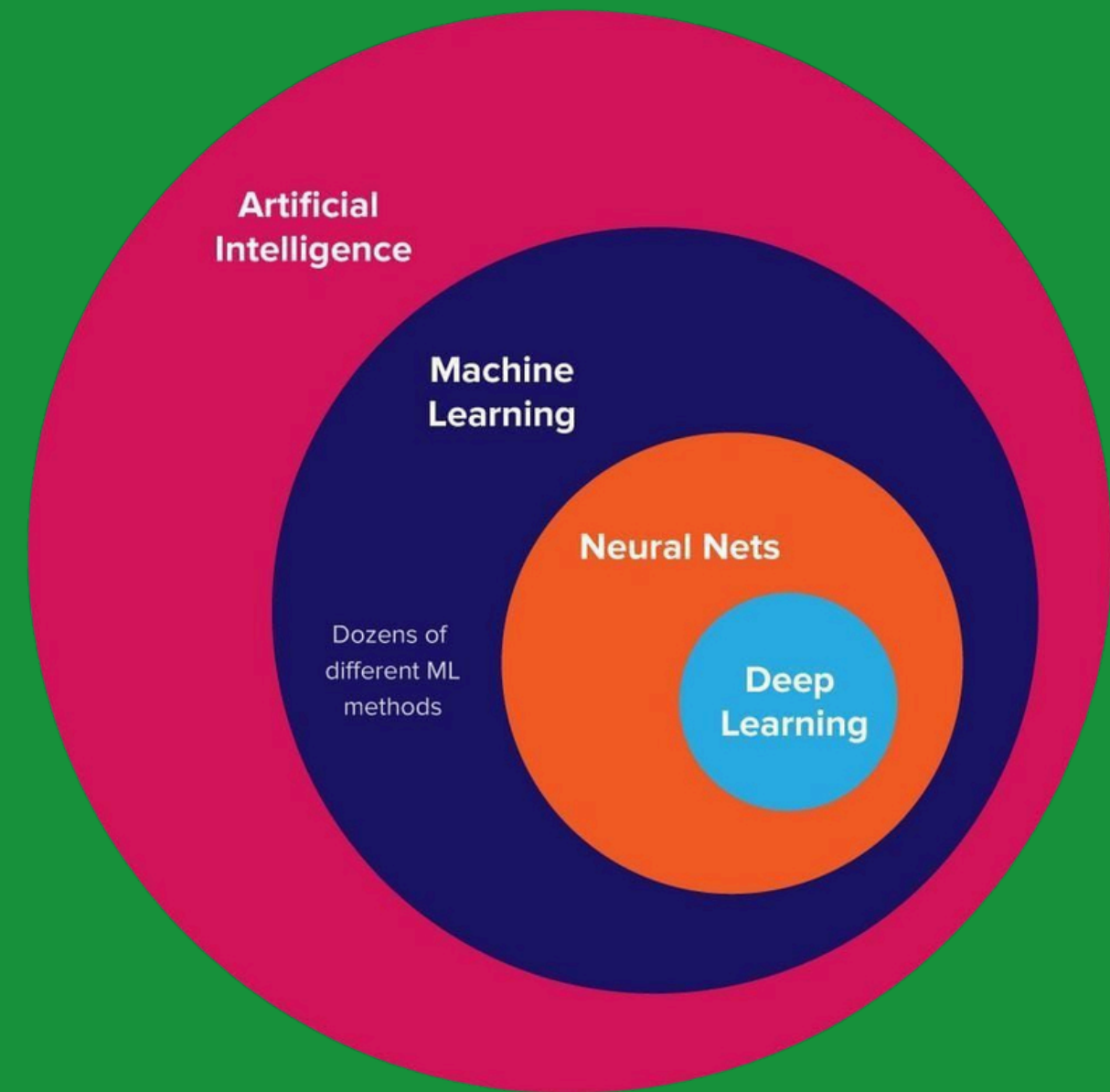
Deep Learning?

LLMs?

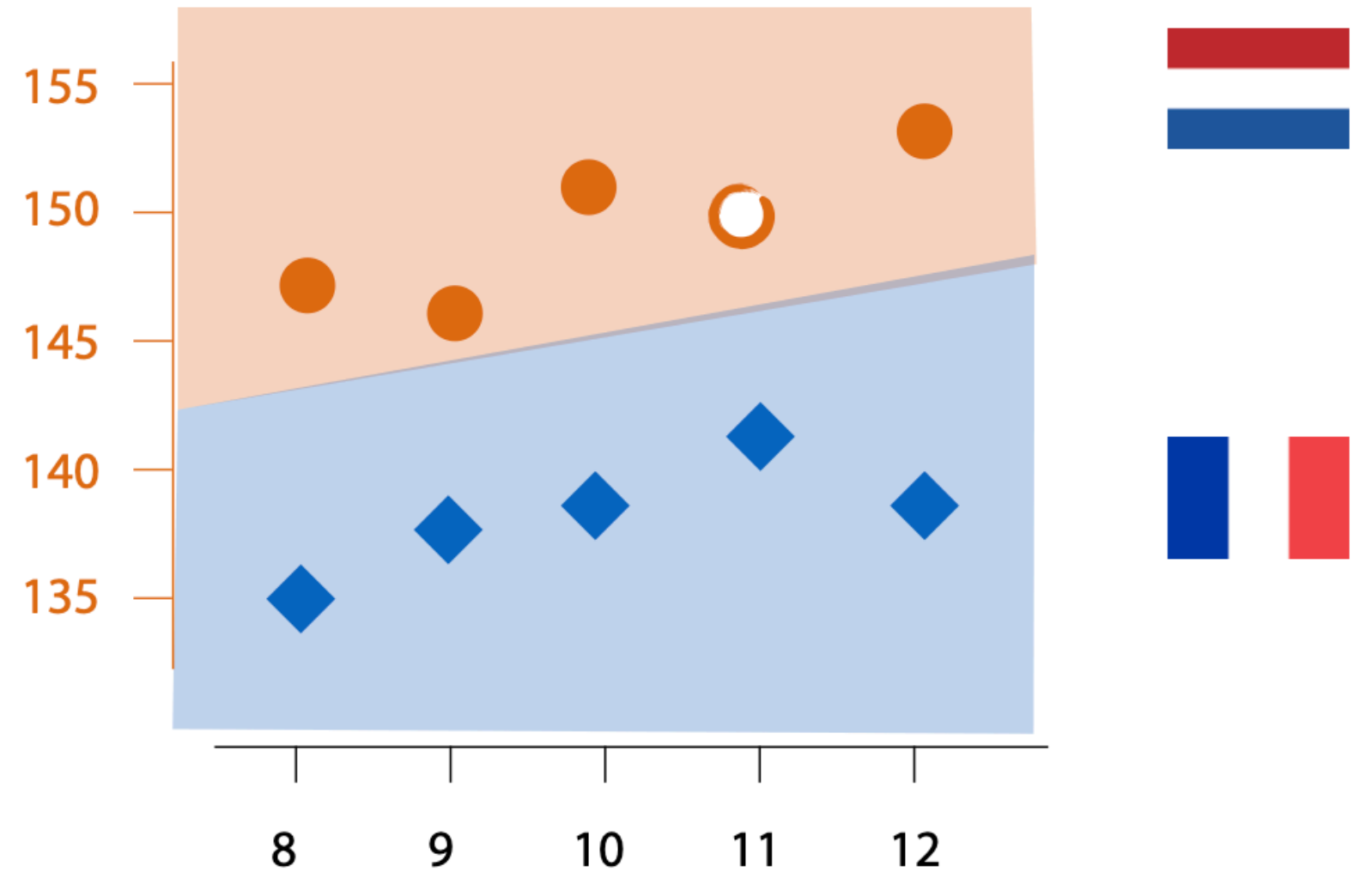
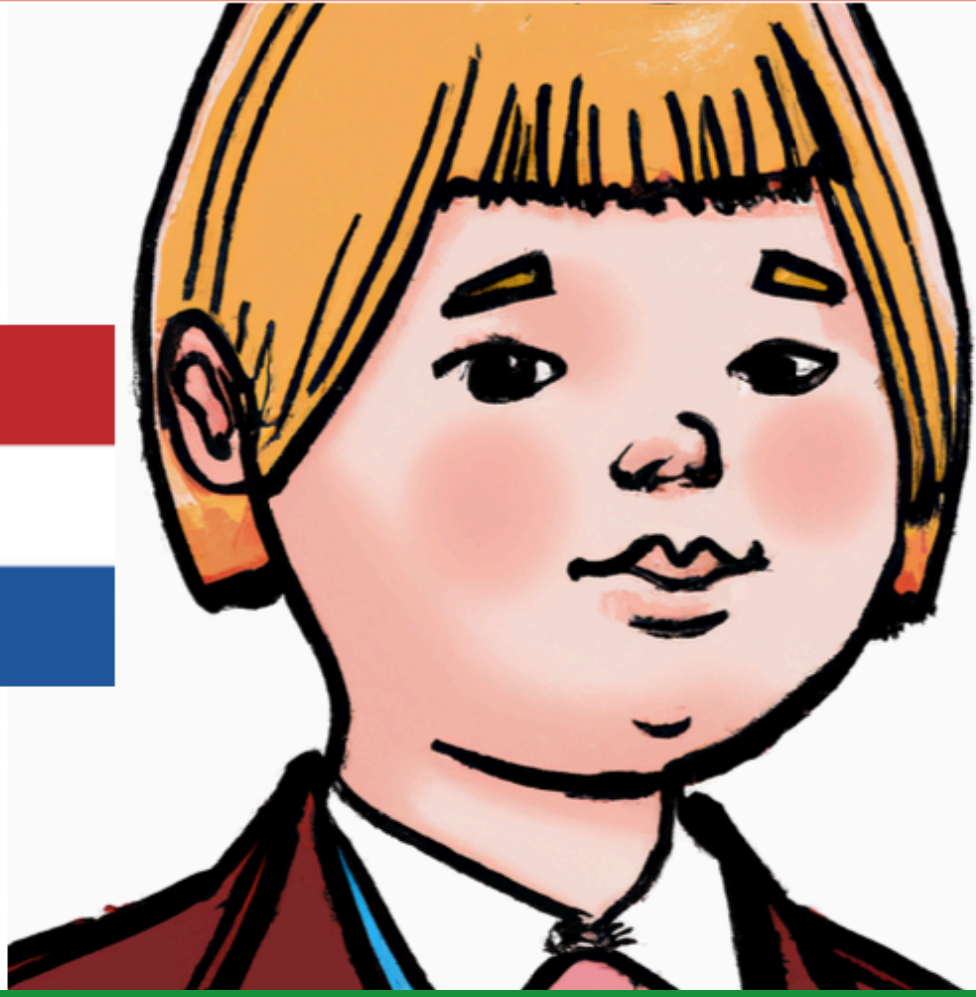


Machine learning ? Artificial Intelligence?

Deep Learning?



What is machine learning?





Epoch
001,386

Learning rate
0.001

Activation
Tanh

Regularization
L1

Regularization rate
0.003

Problem type
Classification

DATA

Which dataset do you want to use?



Ratio of training to test data: 50%

Noise: 0

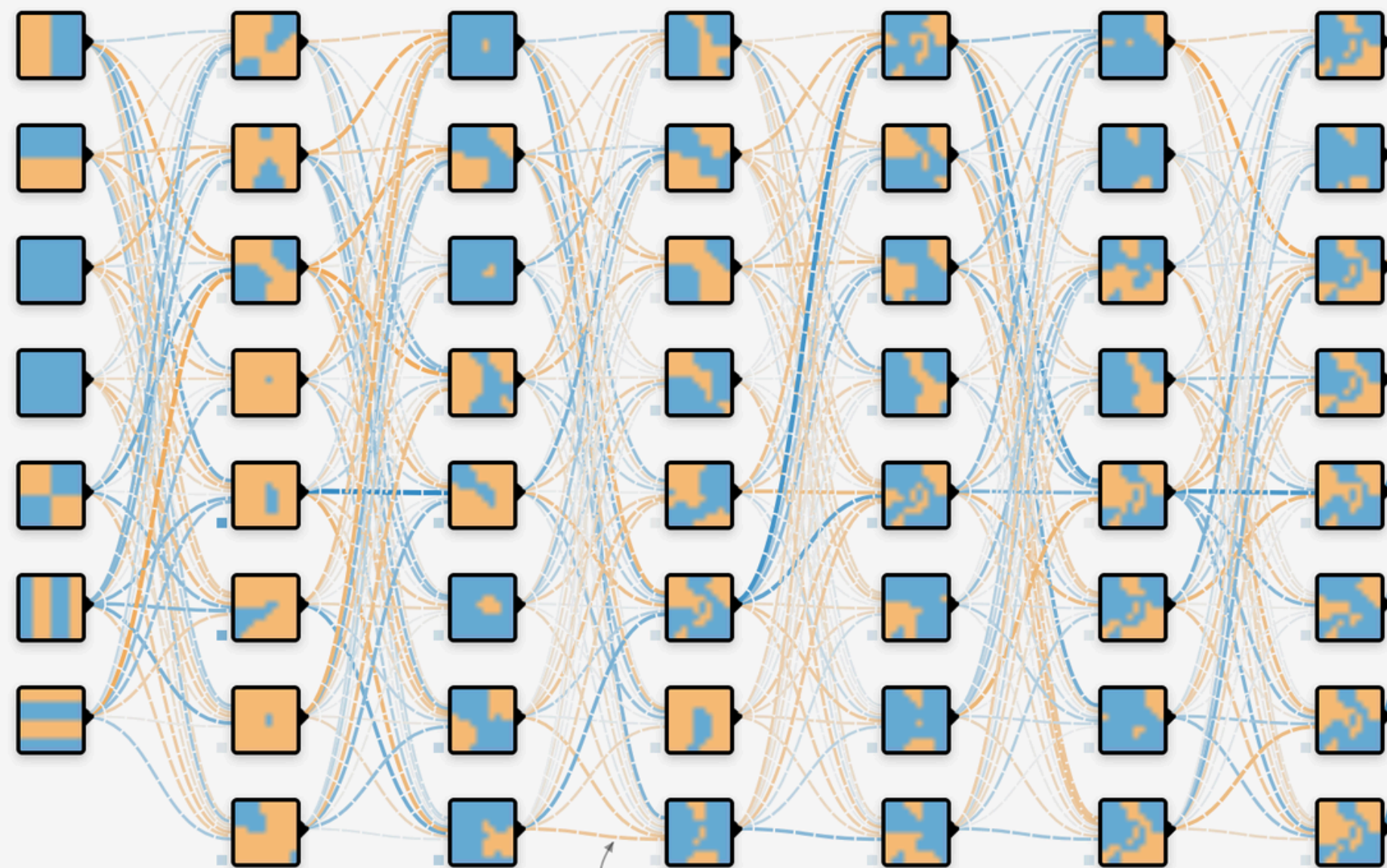
Batch size: 10

REGENERATE

FEATURES

Which properties do you want to feed in?

X_1
 X_2
 X_1^2
 X_2^2
 X_1X_2
 $\sin(X_1)$
 $\sin(X_2)$

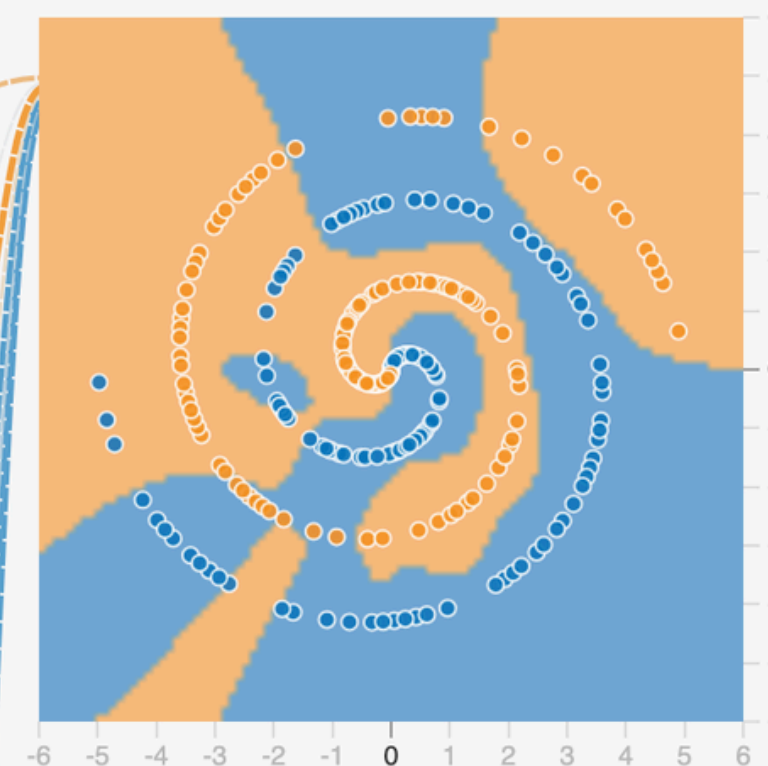


This is the output from one neuron. Hover to see it

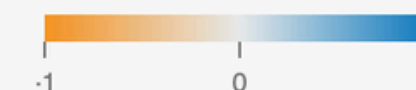
The outputs are mixed with varying weights, shown by the thickness of

OUTPUT

Test loss 0.268
Training loss 0.182



Colors shows data, neuron and weight values.



☐ Show test data ☒ Discretize output

What is next token prediction?

the man fell out of the ...

window

aquarium

pool

cycling

Enter text:

One, two, three,

One , two , three ,

3198 11 734 11 1115 11

Prediction

#	probs	next token ID	predicted next token
0	46.44%	1440	four
1	7.48%	290	and
2	7.31%	1936	five
3	2.66%	393	or
4	2.54%	2237	six
5	2.09%	1115	three
6	1.86%	3863	maybe
7	1.62%	345	you
8	1.23%	257	a
9	0.92%	530	one

Is this all LLMs are?



ScienceNews
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ALL TOPICS ▾ HEALTH HUMANS ▾ LIFE ▾

NEWS MATH

AI cracked an Erdős math problem. Now experts want guardrails

The model disproved a famous conjecture, raising questions about trust, credit and access

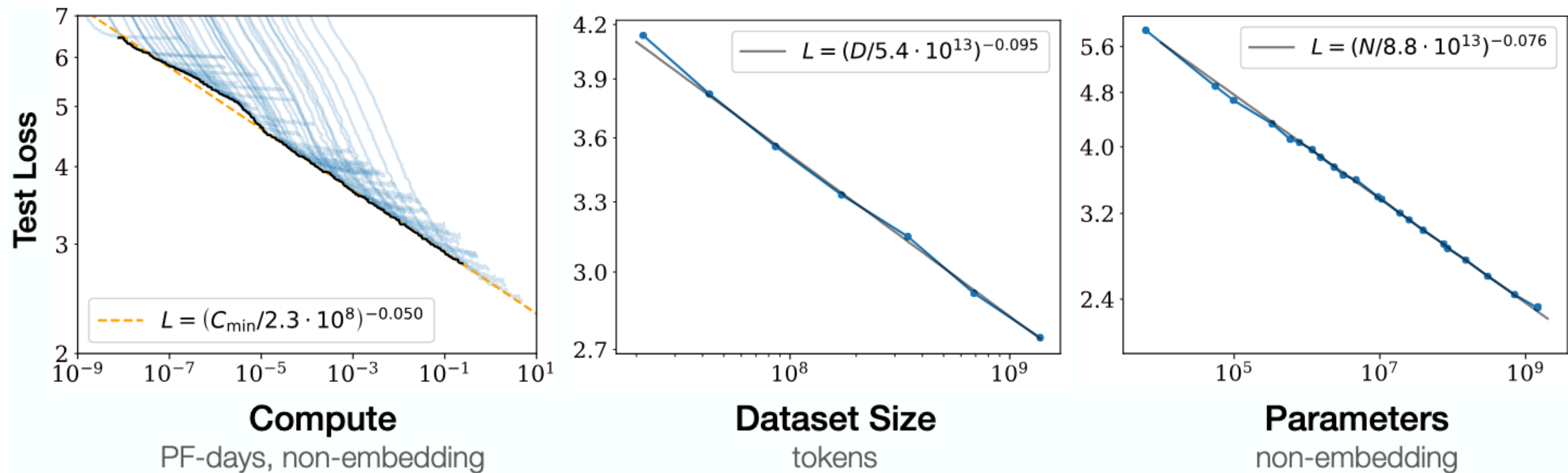


Figure 1 Language modeling performance improves smoothly as we increase the model size, dataset size, and amount of compute² used for training. For optimal performance all three factors must be scaled up in tandem. Empirical performance has a power-law relationship with each individual factor when not bottlenecked by the other two.

Finetuning

- instruction tuning: following instructions
- RLHF: human feedback
- General task solving: math, coding, etc.

CAKE



Comfort Eagle

RONALDLANGESTRAAT **SEARCHING**



What can AI tell us about ourselves?



Prediction in AI and Humans



What is the best way to predict the future?

What is the best way to predict the future?

What is the best way to predict the future?

What is the best way to predict the future?

What is the best way to predict the future?

What is the best way to predict the future?

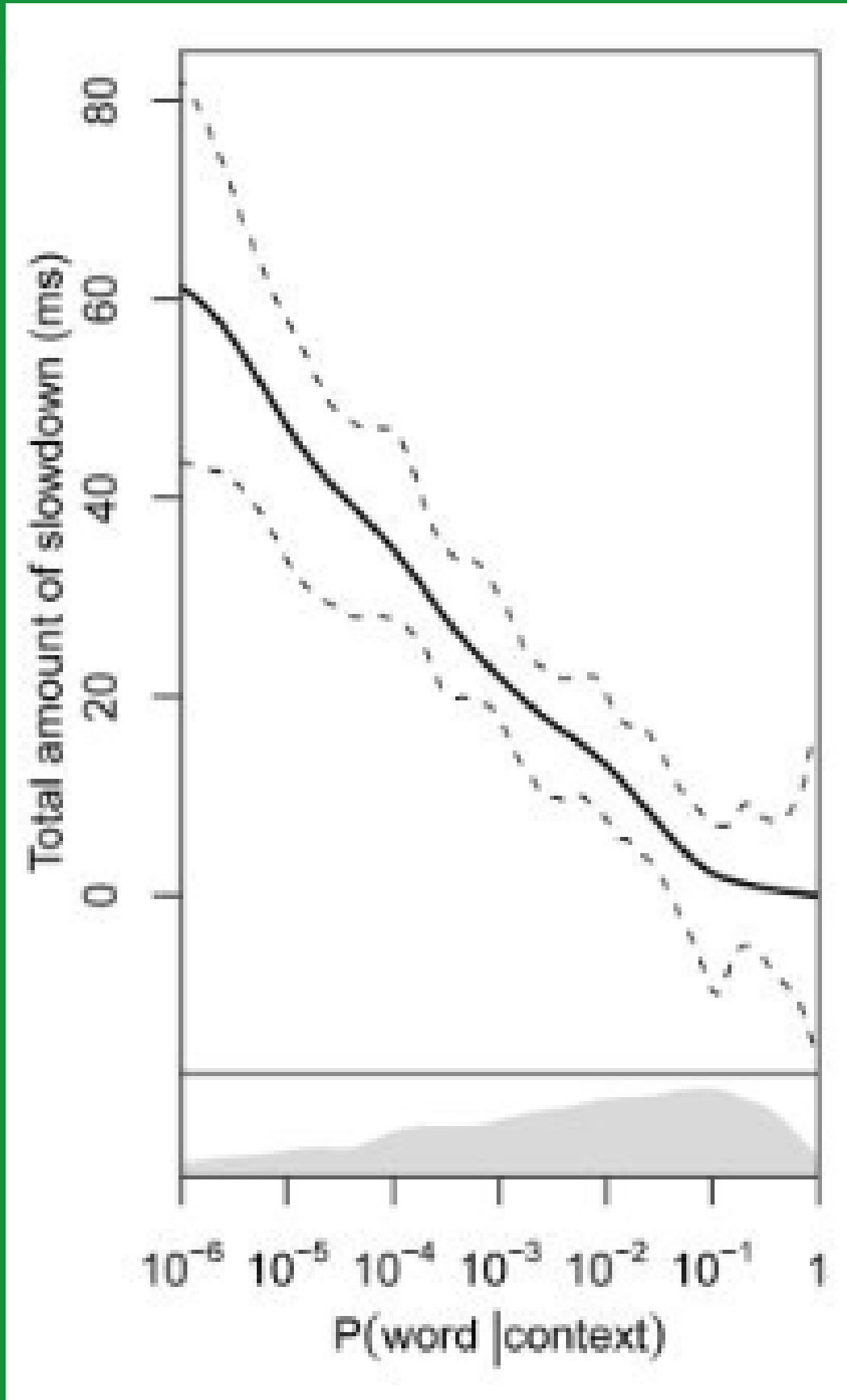
What is the best way to predict the future?

Tracking prediction during reading

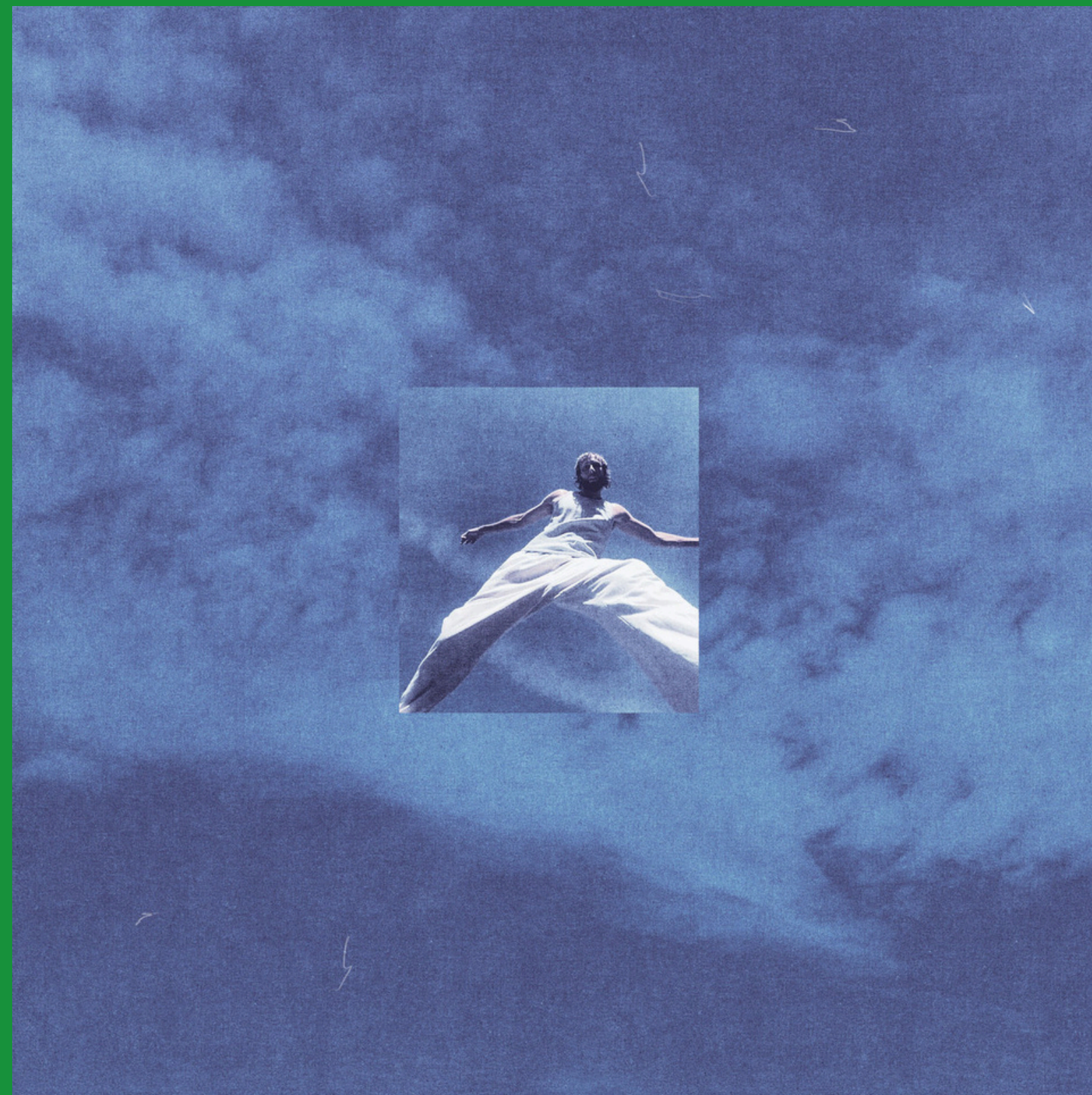
There are advantages and disadvantages of both electronic and hardcopy journals. Hardcopy journals are more easily browsed, more portable and, of course people are very much used to their format. Electronic journals save on paper and their format has improved considerably over the past few years, but there are still problems over managing copyright restrictions and persuading people to use electronic instead of hardcopy journals. There is also the problem of portability. More and more journals are now being published in electronic format, although some publishers will only let you subscribe to an electronic journal provided you also subscribe to the hardcopy (more money for the same thing). Some electronic journals cost over 100% more than their equivalent hardcopy. With all these factors in mind I have been discussing individual and shared-subscriptions with the Biochemistry Department, the RSL and Blackwell's. Whilst I feel that a



Tracking prediction during reading



*How long readers look depends on word **log-probability** (!)*



Your thoughts and questions about AI



Go to wooclap.com

Enter the event
code in the top
banner

Event code

MKAOCJN

Do LLMs “understand” language?

sleeping bag (6.18%)

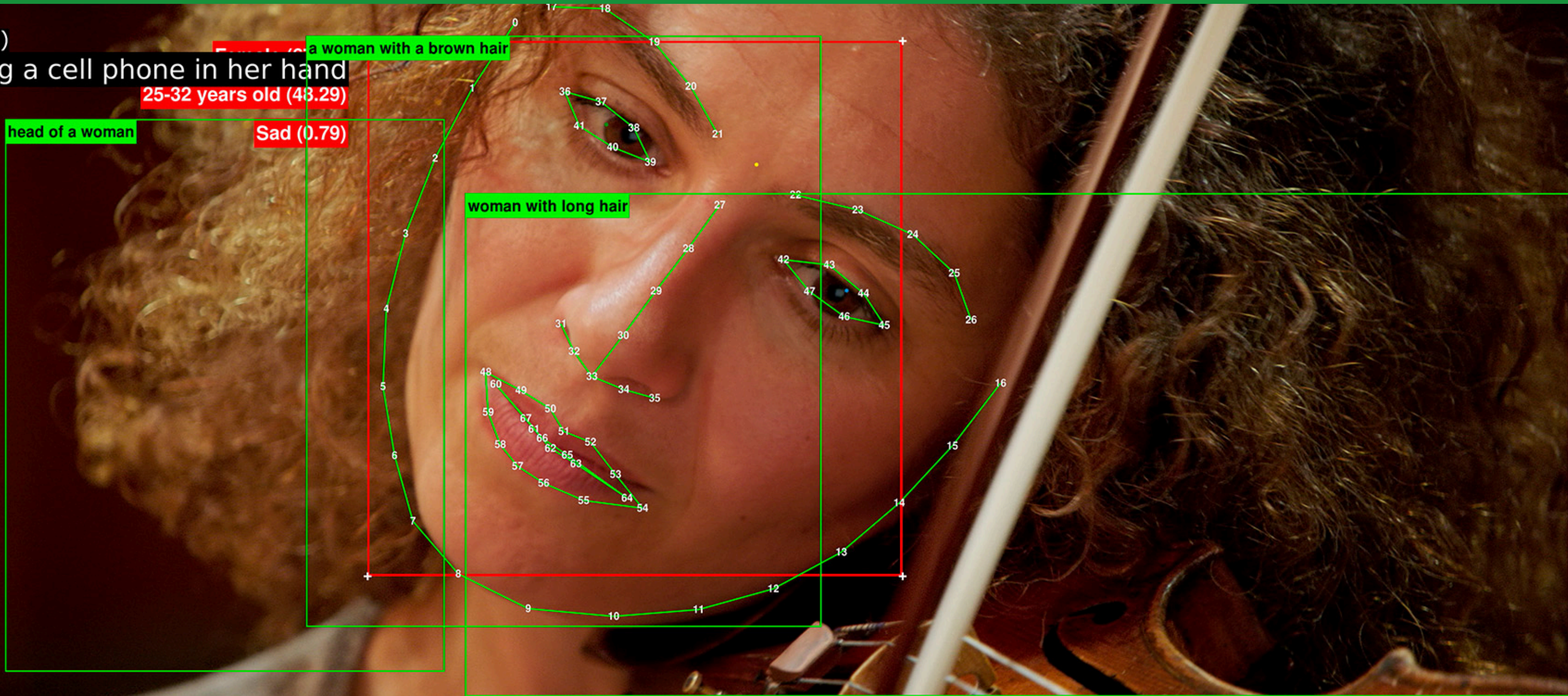
A woman holding a cell phone in her hand

25-32 years old (43.29)

head of a woman

Sad (0.79)

woman with long hair



What does the future look like?





Questions?

Thank you for coming!



Enjoyed the
conversation?

Checkout #33 Can AI
Decode Sound?

w/ Marianne de heer Kloots



Thank you for coming!



AI coding without the
vibes
/ Peter Bloem

